

## SAFETY DATA SHEET

Creation Date 19-November-2010

Revision Date 24-December-2021

Revision Number 6

### 1. Identification

**Product Name** Sodium cyanide

**Cat No. :** S284I-100, S284I-500

**CAS-No** 143-33-9  
**Synonyms** Hydrocyanic acid, sodium salt; Prussiate of soda; Cyanide of sodium

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** CHEMTREC®, Inside the USA: 800-424-9300  
CHEMTREC®, Outside the USA: 001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |            |
|-------------------------------------------------------------|------------|
| <b>Corrosive to metals</b>                                  | Category 1 |
| <b>Acute oral toxicity</b>                                  | Category 1 |
| <b>Acute dermal toxicity</b>                                | Category 1 |
| <b>Acute Inhalation Toxicity</b>                            | Category 1 |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 1 |
| Target Organs - Thyroid.                                    |            |
| <b>Health Hazards Not Otherwise Classified</b>              | Category 1 |
| Contact with acids liberates very toxic gas                 |            |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

May be corrosive to metals  
Fatal if swallowed, in contact with skin or if inhaled  
Causes damage to organs through prolonged or repeated exposure  
Contact with acids liberates very toxic gas

**Precautionary Statements****Prevention**

Take any precaution to avoid mixing with acids  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wear respiratory protection  
Keep only in original container  
Do not get in eyes, on skin, or on clothing  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

Rinse mouth  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER/doctor  
IF ON SKIN: Wash with plenty of soap and water  
Take off immediately all contaminated clothing  
Wash contaminated clothing before reuse  
Absorb spillage to prevent material damage

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in corrosive resistant polypropylene container with a resistant inliner  
Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component      | CAS-No   | Weight % |
|----------------|----------|----------|
| Sodium cyanide | 143-33-9 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

|                                        |                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                        |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                                                     |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                     |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. Systemic Toxicity: Respiratory disorders: Symptoms may include tightness in the chest, flushing, headache, nausea, vomiting, respiratory depression, weakness, irregular heartbeat, abdominal pain, convulsions, and shock: May cause cyanosis (bluish discoloration of skin due to deficient oxygenation of the blood): Exposure may result in death |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                              |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Non-combustible. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Hydrogen cyanide (hydrocyanic acid).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 4             | 0                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.                                                         |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation.                                                                                                                                                            |

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep locked up. Keep away from acids. Keep away from combustible material. Do not store in aluminum containers. Incompatible Materials. Acids. Strong oxidizing agents. Carbon dioxide (CO<sub>2</sub>). Metals.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component      | Alberta                              | British Columbia                     | Ontario TWAEV                    | Quebec                                                   | ACGIH TLV                            | OSHA PEL                              | NIOSH IDLH                                                                     |
|----------------|--------------------------------------|--------------------------------------|----------------------------------|----------------------------------------------------------|--------------------------------------|---------------------------------------|--------------------------------------------------------------------------------|
| Sodium cyanide | Ceiling: 5 mg/m <sup>3</sup><br>Skin | Ceiling: 5 mg/m <sup>3</sup><br>Skin | CEV: 5 mg/m <sup>3</sup><br>Skin | Ceiling: 10 ppm<br>Ceiling: 11 mg/m <sup>3</sup><br>Skin | Ceiling: 5 mg/m <sup>3</sup><br>Skin | (Vacated) TWA:<br>5 mg/m <sup>3</sup> | IDLH: 25 mg/m <sup>3</sup><br>Ceiling: 4.7 ppm<br>Ceiling: 5 mg/m <sup>3</sup> |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.35 mm         | As tested under EN374-3                                |
| Viton (R)      | > 480 minutes     | 0.5mm           | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do

not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                             |
|----------------------------------------|-----------------------------|
| Physical State                         | Solid                       |
| Appearance                             | White                       |
| Odor                                   | bitter almonds              |
| Odor Threshold                         | No information available    |
| pH                                     | 11-12 20 g/l aq. sol        |
| Melting Point/Range                    | 562 °C / 1043.6 °F          |
| Boiling Point/Range                    | 1497 °C / 2726.6 °F         |
| Flash Point                            | No information available    |
| Evaporation Rate                       | Not applicable              |
| Flammability (solid,gas)               | No information available    |
| Flammability or explosive limits       |                             |
| Upper                                  | No data available           |
| Lower                                  | No data available           |
| Vapor Pressure                         | 1 hPa @ 817 °C              |
| Vapor Density                          | Not applicable              |
| Specific Gravity                       | No information available    |
| Bulk Density                           | 750 - 950 kg/m <sup>3</sup> |
| Solubility                             | No information available    |
| Partition coefficient; n-octanol/water | No data available           |
| Autoignition Temperature               | No information available    |
| Decomposition Temperature              | No information available    |
| Viscosity                              | Not applicable              |
| Molecular Formula                      | C N Na                      |
| Molecular Weight                       | 49                          |

## 10. Stability and reactivity

|                                  |                                                                           |
|----------------------------------|---------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                       |
| Stability                        | Stable under normal conditions. Hygroscopic.                              |
| Conditions to Avoid              | Incompatible products. Exposure to moist air or water.                    |
| Incompatible Materials           | Acids, Strong oxidizing agents, Carbon dioxide (CO <sub>2</sub> ), Metals |
| Hazardous Decomposition Products | Nitrogen oxides (NO <sub>x</sub> ), Hydrogen cyanide (hydrocyanic acid)   |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                  |
| Hazardous Reactions              | None under normal processing.                                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component      | LD50 Oral                  | LD50 Dermal                    | LC50 Inhalation              |
|----------------|----------------------------|--------------------------------|------------------------------|
| Sodium cyanide | LD50 = 5.733 mg/kg ( Rat ) | LD50 = 14.602 mg/kg ( Rabbit ) | LC50 = 0.16 mg/L ( Rat ) 1 h |

Toxicologically Synergistic Products No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component      | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------|----------|------------|------------|------------|------------|------------|
| Sodium cyanide | 143-33-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Not mutagenic in AMES Test

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** Thyroid

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Systemic Toxicity: Respiratory disorders: Symptoms may include tightness in the chest, flushing, headache, nausea, vomiting, respiratory depression, weakness, irregular heartbeat, abdominal pain, convulsions, and shock: May cause cyanosis (bluish discoloration of skin due to deficient oxygenation of the blood): Exposure may result in death

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component      | Freshwater Algae | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                             | Microtox   | Water Flea |
|----------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|
| Sodium cyanide | Not listed       | LC50: 0.0558 - 0.0586 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.0391 - 0.0548 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 0.15 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.0712 - 0.0936 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 0.17 mg/L, 96h static (Pimephales promelas)<br>LC50: 0.066 - 0.0852 mg/L, 96h flow-through (Lepomis macrochirus) | Not listed | Not listed |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component      | log Pow |
|----------------|---------|
| Sodium cyanide | -0.44   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                 | RCRA - U Series Wastes | RCRA - P Series Wastes  |
|---------------------------|------------------------|-------------------------|
| Sodium cyanide - 143-33-9 | -                      | not otherwise specified |

### 14. Transport information

#### DOT

UN-No UN1689  
 Proper Shipping Name SODIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

#### TDG

UN-No UN1689  
 Proper Shipping Name SODIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

#### IATA

UN-No UN1689  
 Proper Shipping Name SODIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

#### IMDG/IMO

UN-No UN1689  
 Proper Shipping Name SODIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Subsidiary Hazard Class P  
 Packing Group I

### 15. Regulatory information

#### International Inventories

| Component      | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Sodium cyanide | 143-33-9 | X   | -    | X    | ACTIVE                                        | 205-599-4 | -      | -   |

| Component      | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------|----------|-------|----------|------|------|------|------|-------|-------|
| Sodium cyanide | 143-33-9 | X     | KE-31401 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant | Canadian Environmental | Canada's Chemicals Management |
|-----------|-----------------------------|------------------------|-------------------------------|
|-----------|-----------------------------|------------------------|-------------------------------|

|                | Release Inventory (NPRI)  | Protection Agency (CEPA)<br>- List of Toxic Substances | Plan (CEPA) |
|----------------|---------------------------|--------------------------------------------------------|-------------|
| Sodium cyanide | Part 1, Group A Substance |                                                        |             |

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component      | REACH (1907/2006) - Annex XIV -<br>Substances Subject to<br>Authorization | REACH (1907/2006) - Annex XVII -<br>Restrictions on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 - Candidate<br>List of Substances of Very High<br>Concern (SVHC) |
|----------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Sodium cyanide | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                  | -                                                                                                              |

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component      | CAS-No   | OECD HPV | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|----------------|----------|----------|---------------------------------|------------------------------|--------------------------------------------------|
| Sodium cyanide | 143-33-9 | Listed   | Not applicable                  | Not applicable               | Not applicable                                   |

| Component      | CAS-No   | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|----------------|----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Sodium cyanide | 143-33-9 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Annex I - Y33                         |

## 16. Other information

## Prepared By

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Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

## Creation Date

19-November-2010

## Revision Date

24-December-2021

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24-December-2021

## Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**